

Autonomous Surface Robot for Depth-Resolved Environmental Data

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Problem Statement

The difference in luminosity and temperature at different depths of water is an important metric for determining the success of underwater ecosystems, as energy comes directly from sunlight, and turbid, muddy waters could have a steeper decrease in luminosity as depth increases. The robot aimed to deploy a payload that allowed for detection and analysis of the **temperature and luminosity at different depths** at Dana Point.

Impact

This project demonstrated that a low-cost autonomous surface robot can successfully collect synchronized, **depth-resolved environmental data** in a real coastal environment. It also showed that light attenuation trends can be measured on site, but data quality is strongly dependent on sensor robustness and response time in harsh conditions.

Future Work

If redeployed at Dana Point, there would be priority in improving the sealing of electrical connections, validating sensor performance in saltwater conditions, and optimizing sampling timing for more **stable** and accurate readings. These improvements would allow the robot to more reliably assess factors affecting light attenuation.

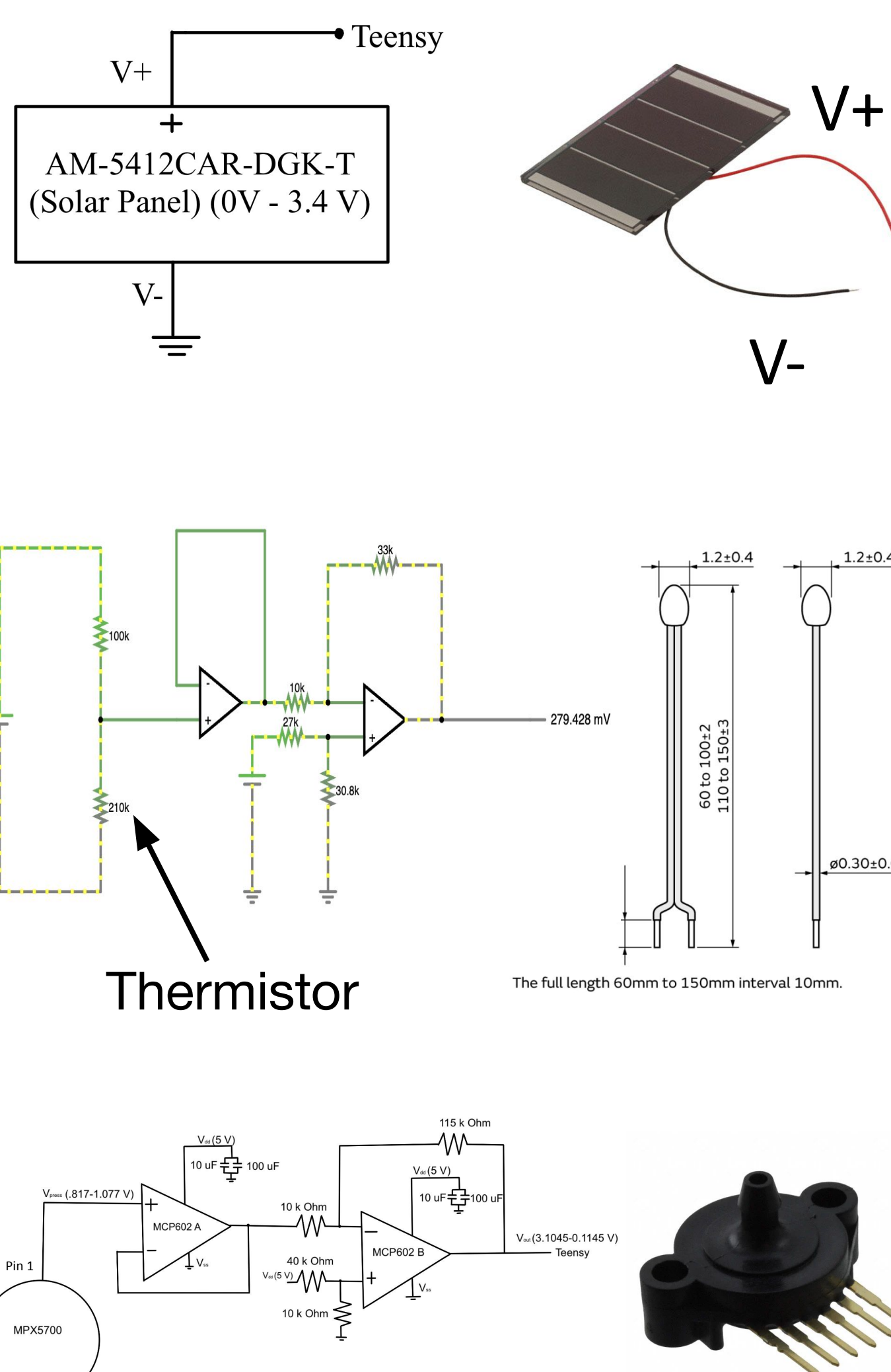
Autonomous Surface Vehicle (ASV) Features

Electrical Schematic & Sensors

Solar sensor ranges from **0 to 100,000 Lux** and outputs 0V to 3.4V. The Panasonic AM-5412CAR-DGK-T amorphous silicon solar cell was chosen because it produces a measurable **voltage proportional to incident light** with a large sensing area. The solar panel V+ was connected directly to the Teensy analog pin A0, and V- to ground.

Temperature sensor (NXFT15WB473FA2B150) is 200kΩ NTC Thermistor, where higher temperature **decreases** resistance. The thermistor is the bottom resistor in the voltage divider, the voltage decreases as temperature increases. With some safety factors, at **20.0°C (120.0kΩ)**, 0.46V and at **10.0°C (200.kΩ)** 2.46 V is outputted. The inputs goes through a unity gain, then an inverting offset amplifier to shift up and amplify the voltage.

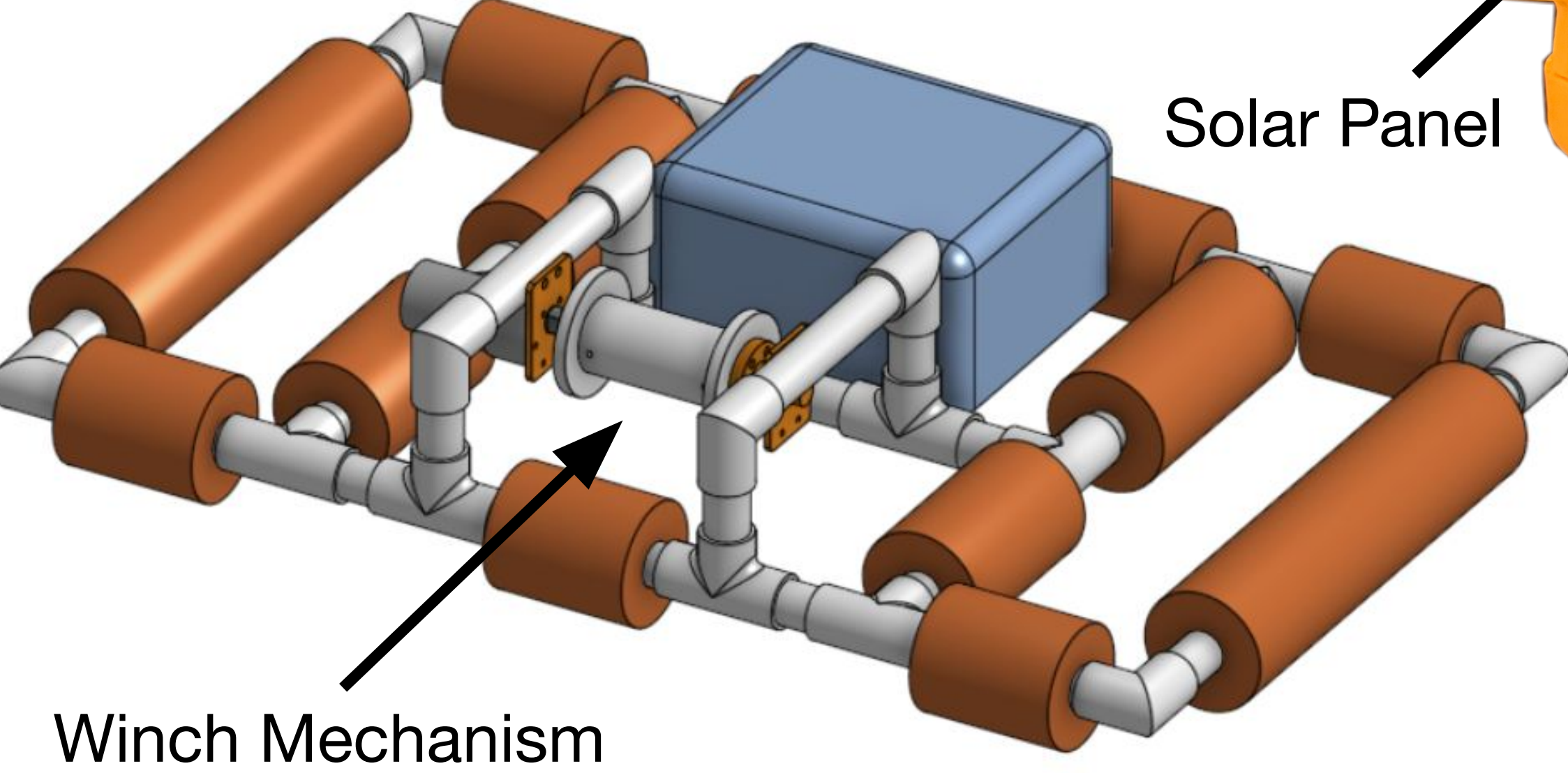
The **pressure** sensor (MPX5700) was prepared for a range of **0 to 4 meters** of depth, which resulted in an expected output of .817 - 1.077 V. Using a unity gain buffer and an inverting offset operational amplifier, outputs were between .1145 and 3.1045 V, within Teensy Range.



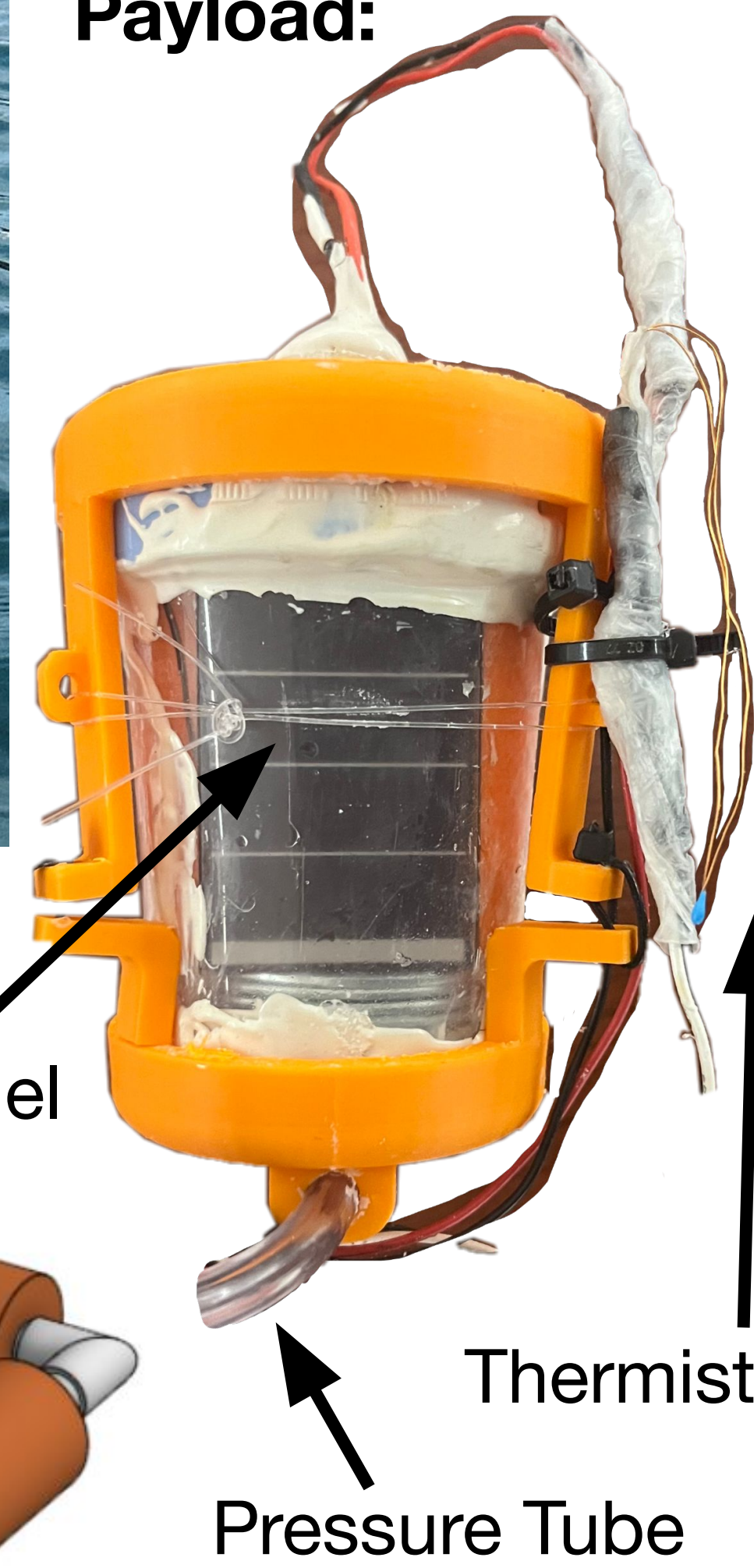
Mechanical Design & Integration



Frame:



Payload:



Experimental Data and Results

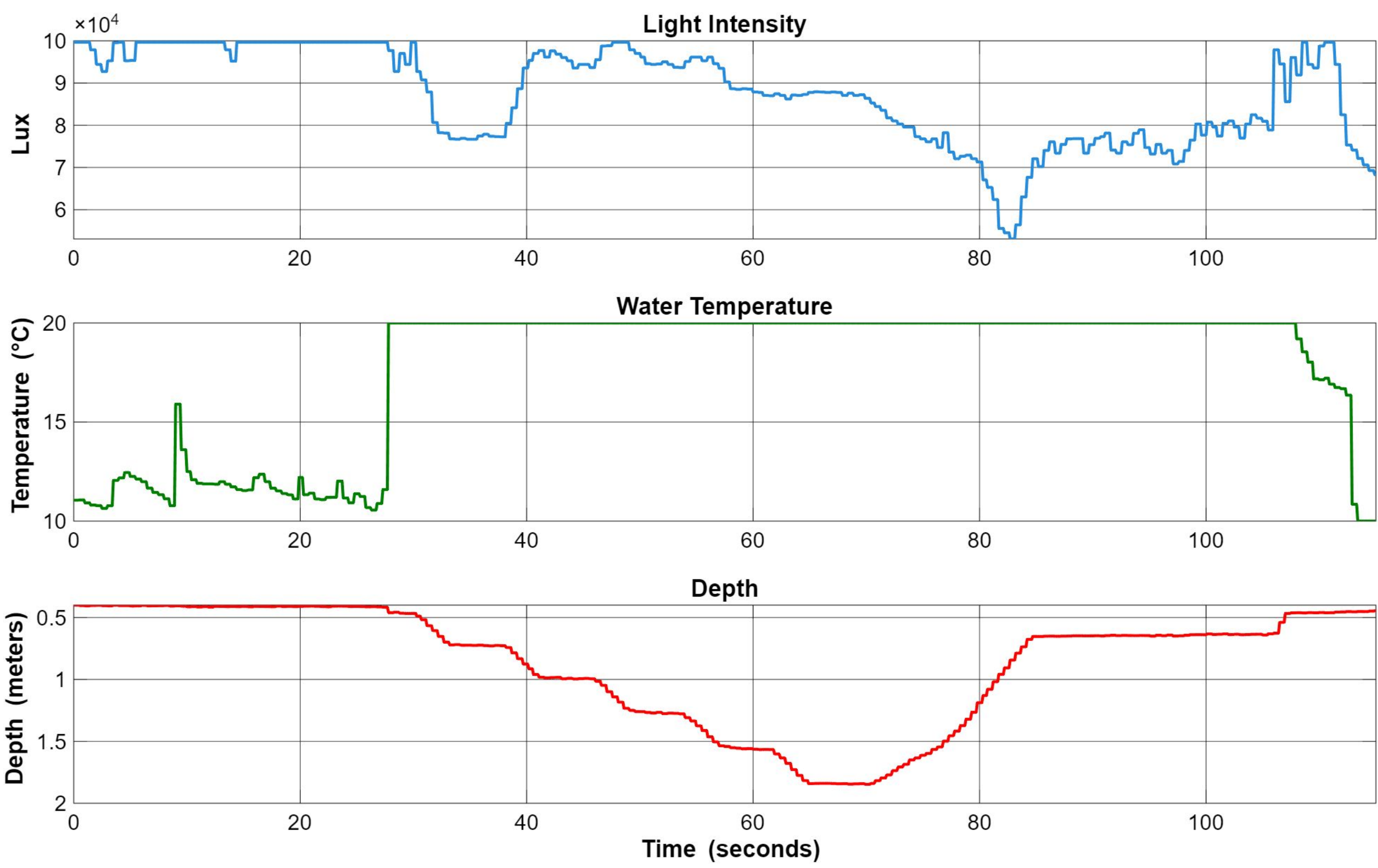
Sensor Calibration

Because the open circuit voltage of the **solar** panel scales **logarithmically** with light rather than linearly, Lux was estimated using an exponential calibration curve: $Lux = Ae(BV)$, where $A = 10.0$ and $B = 2.79$. These constants were found by modifying them until the curve matched the expected lux reading in outdoors sunlight conditions. **Temperature** was designed for a 10.0°C to 20.0°C underwater range. Six distinct baths created a **linear calibration** curve to **map resistances** and output voltages **to temperatures**. To calibrate the **pressure** sensor we used the pressure sensor with the attached tube taped to a yard stick, and took measurements at surface level, and increments of 1 feet up to 3 feet deep, giving us a **calibration curve** for understanding output voltage data.

Ground Truth Measurements

On launch day, the water was 17.9°C, UV ranged from 3-8, and sensors deployed down to 2 meters. When the **solar panel** reached a maximum of **3.3V**, it yielded an AM-1.5 sunlight of 100,00 Lux. When underwater, the solar panel did not output above 3.3V (rail out). For **temperature**: 10.0°C outputted **1.98V**, 20.0°C yielded **0.46V**. For our **pressure** sensor were **3.297V at surface** level, 3.13V at 1ft or .3m, 2.97V at 2ft or .6m, and **2.8V at 3ft** or .91 m of depth. This served as the basis for our calibration curve.

Results & Conclusions



As **depth increased**, measured **lux decreased**, as less light penetrated below the surface. The magnitude of its decrease stayed consistent across different locations, suggesting the **same** clarity of water for the entire bay. Due to waterproofing issues, the thermistor railed out during deployments; but, during smaller submersions the temperature **decreased** with depth as expected.

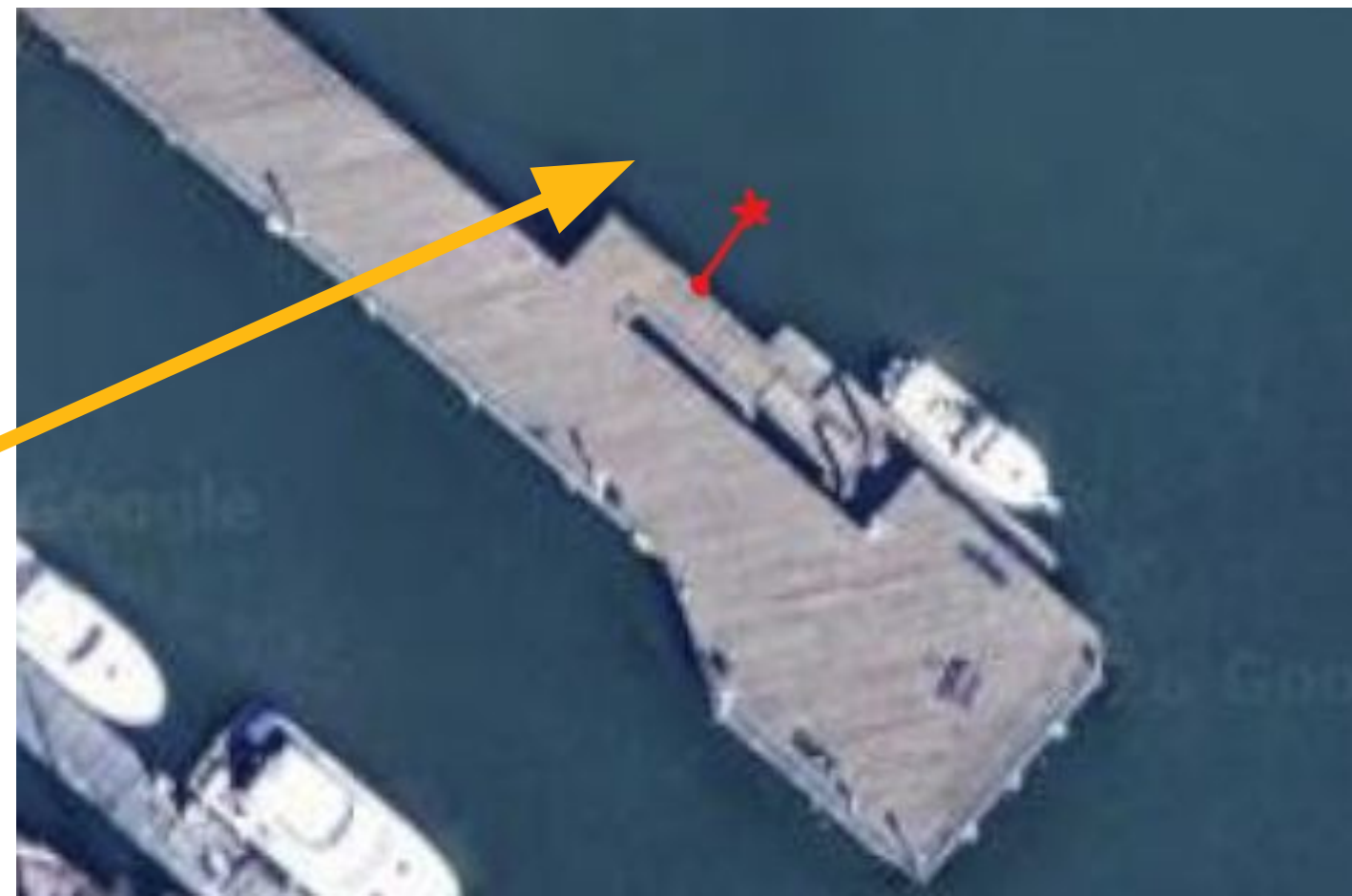
The AUV proved to be a stable and **viable** source of data collection. The **stepped** descent winch programming allowed the sensors to stabilize at discrete depths which can be seen in the depth plot. This shows that low cost sensors can demonstrate information about the state of the bay.

Deployment Details

Location: Dana Point Fishing Pier

Testing Protocol: To prepare for deployment, all sensor wires were first verified to be properly connected to the PCB, followed by sealing the electrical box and placing the robot in the water. The system then traveled outward in a straight line from the deployment point and was safely recovered using a **tether** after completing its measurement run. Life vests were used and trash was left behind.

Dana Point deployment path on April 25, 2026



Acknowledgements

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